

Yash Arora

Department of Mathematics, Indian Institute of Technology – Roorkee, India

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Date of Birth: 11 September, 1997

Fields of Interest

Machine Learning, Support Vector Machines, Deep Learning, Reinforcement Learning

Education

- **Indian Institute of Technology, Roorkee** **Roorkee, India**
Doctor of Philosophy, Mathematics, Course Work, CGPA 9.69/10 Feb 10, 2021–Jan 7, 2026
- **Indian Institute of Technology, Delhi** **New Delhi, India**
Master of Science, Mathematics, First Division, CGPA 8.4/10 2017–2019
- **Deshbandhu College, University of Delhi** **New Delhi, India**
Bachelor of Science (Hons) Mathematics, First Division, 89.69% 2014–2017

Ph.D. Thesis Title

ROBUST VARIANTS OF TWIN SUPPORT VECTOR MACHINES FOR PATTERN CLASSIFICATION WITH APPLICATIONS

Supervisor: Dr. Shiv Kumar Gupta, Professor, Department of Mathematics, IIT Roorkee

Work Experience

- **Genpact India Private Limited** **Bengaluru, India**
Business Analyst 2019–2021
Worked as a Business Analyst in Data Science Team at Genpact.

Research Work

Journal Publications.....

- **Yash Arora**, S. K. Gupta, Intuitionistic fuzzy twin proximal SVM with fuzzy hyperplane and its application in EEG signal classification, **Applied Soft Computing**, 111816, 2024.
doi: <https://doi.org/10.1016/j.asoc.2024.111816>.
- **Yash Arora**, S. K. Gupta, Brain tumor classification using multi-category weighted least square twin SVM with fuzzy hyperplane, **Engineering Applications of Artificial Intelligence** 138, 109450, 2024.
<https://doi.org/10.1016/j.engappai.2024.109450>
- **Yash Arora**, S. K. Gupta, Shuaiyong Li. A novel relative density and non-membership-based intuitionistic fuzzy twin SVM for class imbalance learning, **IEEE Transactions on Fuzzy Systems**, 33 (10), 3795–3807, 2025. <https://doi.org/10.1109/TFUZZ.2025.3605436>.
- **Yash Arora**, Scindhiya Laxmi, S. K. Gupta. A robust twin support vector machine with huberized hinge loss function with an application in handwritten digit recognition, **Pattern Analysis and Applications**, 29 (58), 2026. <https://doi.org/10.1007/s10044-026-01611-4>
- **Yash Arora**, S. K. Gupta, M. Tanveer. A robust multi-view support vector machine with the RoBoSS loss function, **Neural Networks**, 201, 108937, 2026. <https://doi.org/10.1016/j.neunet.2026.108937>

- Nikita Grewal, **Yash Arora**, S. K. Gupta, Sanjeev Kumar. Class probability and polycentric fuzzy least squares twin SVM for imbalanced data, To be appear in **Knowledge and Information Systems**.

Conference Publications.....

- Nikita Grewal, **Yash Arora**, S. K. Gupta, Sanjeev Kumar. Generalized bell fuzzy proximal twin support vector machine for multi-class classification, **FUZZ-IEEE, Reims, France, 2025**, DOI: 10.1109/FUZZ62266.2025.11152260

Publications Under Review.....

- **Yash Arora**, S. K. Gupta. On entropy-based intuitionistic fuzzy least squares twin support vector machine, **Fuzzy Sets and Systems**, *Revision submitted*.

Workshops/Conferences

- Presented a paper entitled "*Generalized bell fuzzy proximal twin support vector machine for multi-class classification*" at the **IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2025)**, held at Reims, France, during 06-09 July, 2025.
- Presented a paper entitled "*Improved intuitionistic fuzzy least square twin support vector machine for class imbalance learning*" at the **34th European Conference on Operational Research (EURO 2025)**, held at University of Leeds, Leeds, UK, during 22-25 June, 2025.
- Presented a paper entitled "*Multi-category weighted least square twin SVM with fuzzy hyperplane*" at the **International Conference on Statistical Learning (ICSL-2025)**, held at IIT Ropar, Punjab, India, during 06-08 March, 2025.
- Attended **The Fifth Indian Symposium on Machine Learning (IndoML 2024)**, held at BITS Pilani Goa Campus, Goa, India, between 21-23 December, 2024.
- Attended a **4th Summer School on Large Scale Optimization** held at IIT Roorkee, India from May 18- May 26, 2024.
- Presented a paper entitled "*Intuitionistic fuzzy twin proximal SVM with fuzzy hyperplane and its application in EEG signal classification*" at the **International Conference on Computations and Data Science (CoDS-2024)**, held at IIT Roorkee, Uttarakhand, India during 8-10 March, 2024.
- Presented a poster entitled "*Intuitionistic fuzzy twin proximal SVM with fuzzy hyperplane and its application in EEG signal classification*" at the **Fourth Indian Symposium on Machine Learning (IndoML 2023)**, held at IIT Bombay, Mumbai, India between 21-23 December, 2023.
- Presented a paper entitled "*Intuitionistic fuzzy twin proximal SVM with fuzzy hyperplane*" at the **10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)**, held at Tokyo, Japan, from August 20 -August 25 , 2023.
- Presented a paper entitled "*Huberized twin support vector machine for binary classification problems*" at the **International Symposium on Applied Optimization and Game Theoretic Models for Decision Making**, organized by Indian Statistical Institute, Delhi from February 01-February 03, 2023.
- Presented a paper entitled "*Huberized twin support vector machine for binary classification problems*" at the **International conference on Optimization, Learning and Analytics in Business (OLAB-2022)**, organized by Department of Mathematics, Heritage Institute of Technology, Kolkata, India and Operation Research Society of India (ORSI) from December 15-December 17, 2022.
- Participated in the workshop titled "*Data Science in Finance*" during the sixth workshop and conference on **Statistical Methods in Finance 2021**, organized jointly by Chennai Mathematical Institute and Indian Statistical Institute, Bangalore from June 27-July 01, 2021.

Academic Achievements

- Awarded CIS travel grant from the IEEE Computational Intelligence Society (CIS) for in-person participation in the "*IEEE International Conference on Fuzzy Systems (FUZZ-IEEE 2025)*", to be held in Reims, France, from 06-09 July, 2025
- Awarded financial assistance from Anusandhan National Research Foundation (ANRF) to participate in the "*34th European Conference on Operational Research (EURO 2025)*", to be held in Leeds, UK from 22-25 June, 2025.
- Received financial support for Registration fee from the "10th International Congress on Industrial and Applied Mathematics, Tokyo, Japan, 2023.
- Qualified the Joint CSIR UGC Test for Eligibility for Assistant Professor in Mathematics in June 2019 with an All India Rank of 78.
- Qualified GATE-Graduate Aptitude Test in Engineering (Mathematics) 2019.
- Qualified IIT JAM-Joint Admission Test for M.Sc. in Mathematics 2017 with an All India Rank of 92.

Teaching Experience

- **Department of Mathematics, IIT Roorkee** **Roorkee, India**
○ *Teaching Assistant* **2022-2024**
Teaching assistant for following courses: MA-503- Operations Research (spring 2024), MAN 001-Mathematics-I (Fall 2023, Fall 2022), MAN 010- Optimization Techniques (Spring 2023) and MAN 002- Mathematical Methods (Spring 2022).
- **National Programme on Technology Enhanced Learning (NPTEL)** **Roorkee, India**
○ *Teaching Assistant* **2022-2025**
Worked as a graduate teaching assistant for the online Swayam-NPTEL courses on Multivariable Calculus (Spring 2024), Essential Mathematics for Machine Learning (Fall 2023, 2024, 2025), Multivariable Calculus (Spring 2023) and Matrix Analysis and its Applications (Fall 2022).
- **Coursera** **Roorkee, India**
○ *Teaching Assistant* **2022-2023**
Worked as a teaching assistant for the online coursera course on Linear Algebra Basics a part of course Post Graduate Certificate in Data Science and Machine Learning .

Technical Skills

- **Languages and Tools:** R, Python, MATLAB, LaTeX.
- **Software Skills:** MS Office: MS Excel, MS Word, MS Powerpoint.

Miscellaneous

- Volunteer:**
- Volunteered at the International Conference on Computations and Data Science (CoDS-2024), India, contributing to technical and management aspects.
- Professional Membership:**
- **Student member of IEEE** **2025-Present**
○ www.ieee.org

References

- Dr. S. K. Gupta, Professor, Department of Mathematics, IIT Roorkee
- Dr. Sanjeev Kumar, Professor, Department of Mathematics, IIT Roorkee